Abel Shine Varghese

abelshinevarghese@gmail.com — +91 7898670716 — linkedin.com/in/abelshinev —
github.com/abelshinev — abelsv-portfolio.vercel.app

Professional Summary

Fullstack Developer and Systems Builder with experience designing and deploying end-to-end web applications and real-time systems. Skilled in building scalable frontend architectures, backend APIs, and production-ready platforms with a focus on performance, user experience, and reliability.
Experienced in taking products from concept to deployment, including system design, data handling, and integration of intelligent features where needed. Strong foundation in React, Next.js, backend systems, and database-driven applications.

Education

Indian Institute of Information Technology, Design & Manufacturing, Jabalpur Aug 2023 – May 2027
B.Tech in Computer Science & Engineering
Relevant Coursework: Machine Learning, Data Structures and Algorithms, Database Management, Agile Software Development, Generative AI

The Emirates National School, Sharjah Sep 2011 – Mar 2023
Higher Secondary Education, CBSE

Skills Summary

Programming: Python, JavaScript (ES6+), SQL, C/C++, Java
Fullstack & Deployment: React, Next.js, TailwindCSS, REST APIs, PostgreSQL
Backend: Node.js, MongoDB, API Design, MongoDB, Database Modeling
Machine Learning & AI: PyTorch, TensorFlow, Scikit-Learn, YOLOv8, Computer Vision, Time-Series Forecasting, Grad-CAM, Feature Engineering
Cloud & Dev: Oracle Cloud Infrastructure (OCI), Git, Agile Development
Spoken Languages: English (Fluent), Malayalam (Native), Hindi, Arabic (A1), German (A2)

Experience

AI-Enabled Brand Developer Jun 2025 – Present
Zeyno Creative Web Agency @ zeyno.my
- Co-founded a digital product and branding agency delivering digital branding and fullstack web solutions.
- Architected and deployed scalable web platforms for 4 clients including a collegiate racing team.
- Designed frontend systems with performance-focused architecture. (NEXT Framework)
- Managed full product lifecycle from requirement gathering to deployment and client delivery.

UI/UX Developer Intern May 2025 – Jul 2025
QBytez Infolabs
- Developed responsive frontend systems using React and Tailwind CSS for a 20-page corporate platform.
- Collaborated with backend engineers to integrate APIs and optimize user workflows.
- Improved responsiveness, accessibility, and page performance across devices.

Selected Projects

VantaBlock — Behavior-Based Agent Detection System HackByte April, 2026
ML + Real-Time Systems + Browser Instrumentation
- Built a behavior-based bot detection system using interaction dynamics instead of static fingerprinting.
- Collected custom dataset from a simulated workflow with human and agent sessions (5kevents/session).
- Engineered behavioral features including inter-event timing variance, mouse trajectory efficiency, reaction delay, and error patterns.
- Identified key signal: agent decision latency (800–2500ms) vs human reaction time (100–800ms).
- Trained ensemble classifier (Decision Tree + Logistic Regression) for segment-level prediction.
- Designed incremental scoring system aggregating predictions across interaction segments.
- Deployed FastAPI inference server with real-time predictions via Chrome extension (Plasmo).
- Implemented client-side enforcement layer to block detected agents dynamically.

Explainable Airport Baggage Threat Detection System Jan 2026 – March 2026
Computer Vision (YOLOv8) + Explainable AI

- Built a multi-class X-ray baggage threat detection system using the PIDray dataset (75k+ images).
- Converted COCO annotations into YOLO format and trained a YOLOv8 model for real-time detection.
- Detected 12 prohibited item categories under real-world clutter and occlusion conditions.
- Leveraged 42,000+ negative samples to significantly reduce false positives.
- Achieved 92% recall while maintaining high precision across training epochs.
- Integrated Grad-CAM for visual explainability, highlighting model attention regions for safety-critical transparency.

Intelligent Food Spoilage Monitoring System Jan 2026 – March 2026
IoT + Computer Vision + Time-Series Modeling

- Designed and fabricated an end-to-end spoilage detection system integrating computer vision and embedded ML.
- Developed a dual-stage freshness engine extracting HSLA features from chemical indicators inside refrigeration units.
- Built a time-series progression model using rate-of-change metrics to predict spoilage percentage and remaining safe time.
- Deployed inference pipeline on Raspberry Pi for edge-based monitoring.
- Integrated a Flutter companion app enabling closed-loop user feedback for adaptive threshold refinement.

Fit Forecaster – Time Series Traffic Prediction System Dec 2024 – Aug 2025
Ensemble ML Model

- Developed an ensemble forecasting system combining Random Forest, Linear Regression, and XGBoost.
- Engineered lag features, scaling pipelines, and cross-validation workflows for stable predictions.
- Achieved 92.6% accuracy in predicting gym congestion across workout categories.
- Built a Streamlit-based visualization dashboard to interpret predictions and usage patterns.

Certifications

Oracle Cloud Infrastructure Generative AI Professional – Oracle Sep 2025
Physics-Aware Deep Learning – GIAN, IIITDM Jabalpur Feb 2025
Front End Engineer Career Path – Codecademy Jan 2025
Python Programming Masterclass – Udemy Aug 2024